Stokes' theorem (16.8)

Recall: If $\vec{F} = (P, Q, R)$, then:

$$\operatorname{div}(\vec{F}) = \nabla \cdot \vec{F} = P_x + Q_y + R_z$$

$$\operatorname{curl}(\vec{F}) = \nabla \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & Q & R \end{vmatrix}$$

$$= (R_y - Q_z,\ P_z - R_x,\ Q_x - P_y)$$

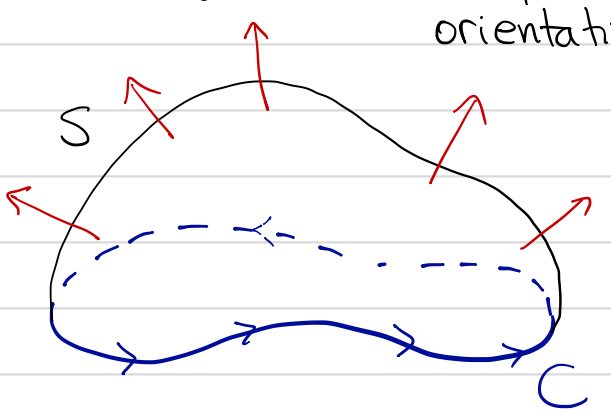
Stokes' Theorem

$$\boxed{\int_C \vec{F} \cdot d\vec{r} = \iint_S \operatorname{curl}(\vec{F}) \cdot d\vec{S}}$$

$\vec{F}$ vector field

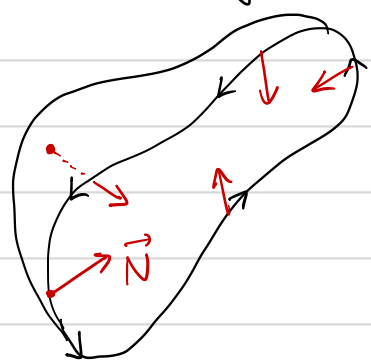
S surface } with compatible orientations

C boundary curve

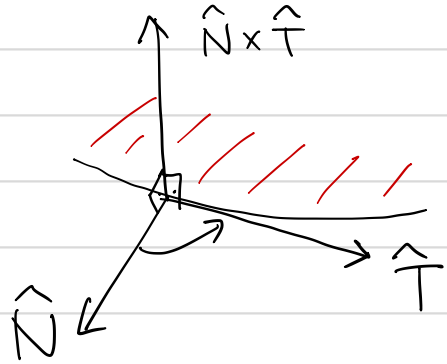


Orientation of C:

Walk along C, head in direction of $\vec{N}$ $\rightsquigarrow$ surface to your left.

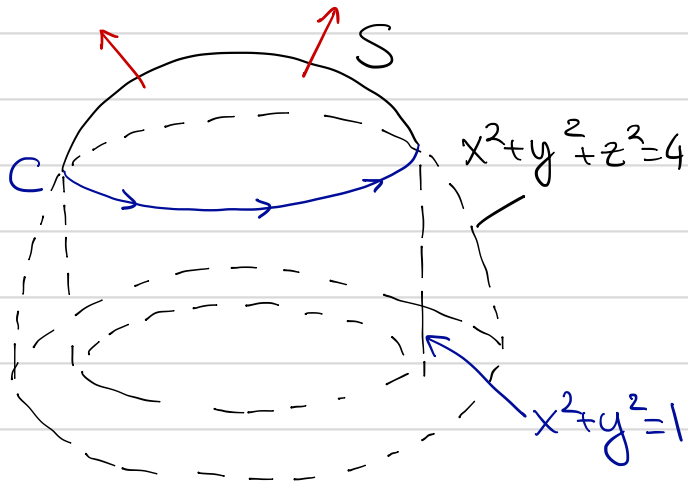


Right-hand rule:



$\hat{N} \times \hat{T}$ points <u>into</u> the surface.

Example: $\vec{F} = \langle 0, xz, xy \rangle$



$$\iint_S \nabla \times \vec{F} \cdot d\vec{S} = \int_C \vec{F} \cdot d\vec{r} = \sqrt{3}\pi$$

On C: $\underbrace{x^2+y^2}_{1} + z^2 = 4$

$\Rightarrow z = \sqrt{3}$ $\qquad dz = 0$

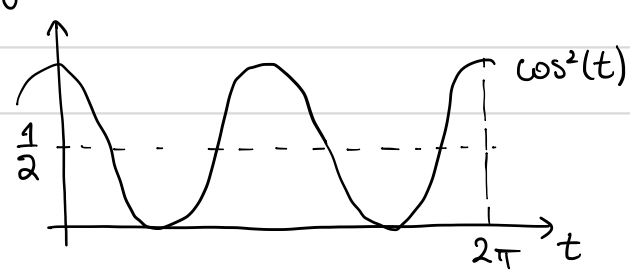
$x = \cos(t)$ $\qquad dx = -\sin(t)\,dt$

$y = \sin(t)$ $\qquad dy = \cos(t)\,dt$

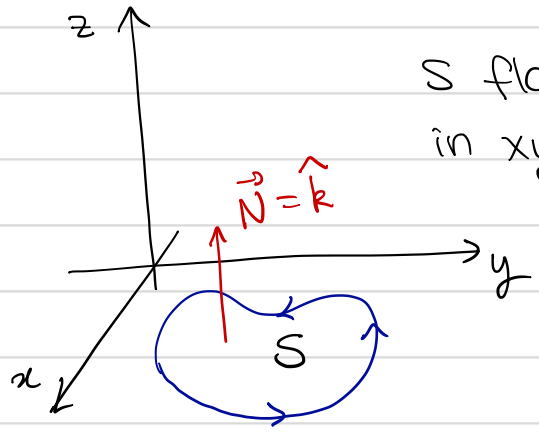
$$\int_C \vec{F} \cdot d\vec{r} = \int_C \langle 0, xz, xy \rangle \cdot \langle dx, dy, dz \rangle$$

$$= \int_0^{2\pi} \langle 0, \sqrt{3}\cos t, \cos t \sin t \rangle \cdot \langle -\sin t, \cos t, 0 \rangle \, dt$$

$$= \int_0^{2\pi} \sqrt{3} \cos^2 t \, dt = \boxed{\sqrt{3}\pi}$$



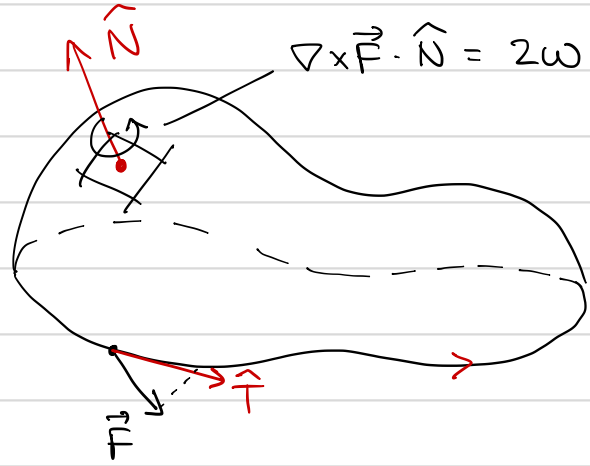
Relationship with Green's theorem



S flat, in xy-plane:

$$\int_C \vec{F} \cdot d\vec{r} = \iint_S \underbrace{(\nabla \times \vec{F}) \cdot \hat{k}}_{Q_x - P_y} \, dA$$

$\vec{F}$ = Fluid velocity ← Skip in class



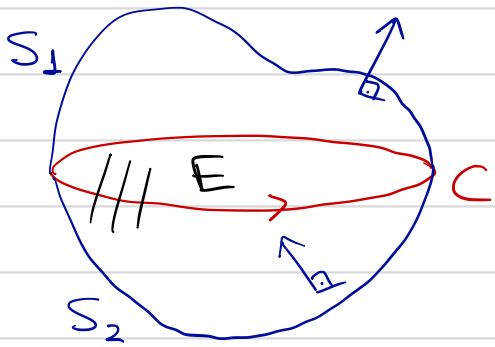
$$\iint_S \nabla \times \vec{F} \cdot \underbrace{\hat{N}\, dS}_{d\vec{S}} = \int_C \vec{F} \cdot \underbrace{\hat{T}\, ds}_{d\vec{r}}$$

$\nabla \times \vec{F} \cdot \hat{N} = 2\omega$

$= 2$(normal component of angular velocity)

$\vec{F} \cdot \hat{T}$ = tangential component of velocity.

Surface indepence in Stokes:



$$\underbrace{\iint_{S_1} \nabla\times\vec{F}\cdot\hat{N}\,dS}_{I_1} = \int_C \vec{F}\cdot d\vec{r}$$

$$= \underbrace{\iint_{S_2} (\nabla\times\vec{F})\cdot\hat{N}\,dS}_{I_2}$$

Another way to get that $I_1 = I_2$:

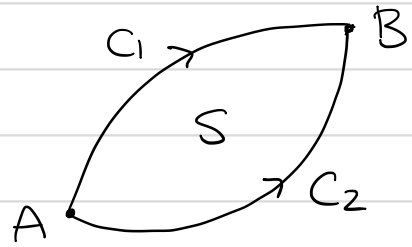
$$I_1 - I_2 = \iint_{S_1 - S_2} \nabla\times\vec{F}\cdot\hat{N}\,dS$$

$$= \iiint_E \underbrace{\nabla\cdot(\nabla\times\vec{F})}_{=0 \text{ for any } \vec{F}}\,dV = 0$$

Divergence theorem

E = solid region enclosed by S_1 & S_2.

Curve independence in FTC:



$$\underbrace{\int_{C_1} \nabla f\cdot d\vec{r}}_{J_1} = f(B) - f(A)$$

$$= \underbrace{\int_{C_2} \nabla f\cdot d\vec{r}}_{J_2}$$

Another way to get $J_1 = J_2$:

$$J_1 - J_2 = \int_{C_1 - C_2} \nabla f\cdot d\vec{r}$$

$$\overset{\text{Stokes}}{=} \iint_S \underbrace{\nabla\times(\nabla f)}_{=0}\cdot d\vec{S}$$

Supplementary notes

- Proof of Stokes' theorem

$$\boxed{\oint_C \vec{F}\cdot d\vec{r} = \iint_S \nabla\times\vec{F}\cdot d\vec{S}} \quad (*)$$

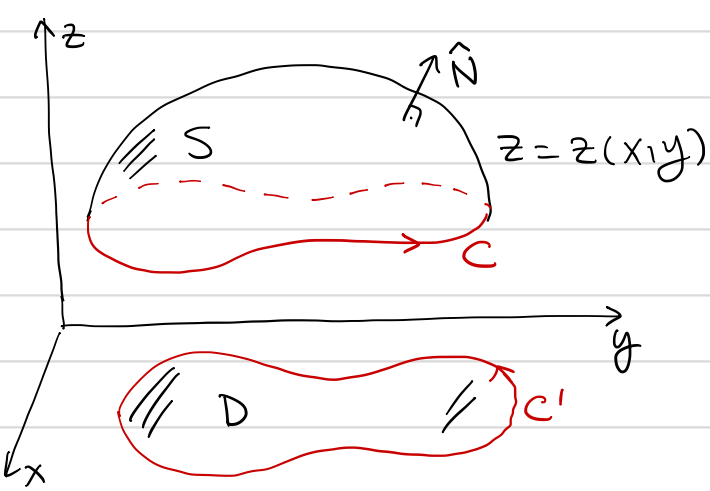
Because

$$\nabla\times(\vec{F}+\vec{G}) = \nabla\times\vec{F} + \nabla\times\vec{G},$$

it suffices to check that $(*)$ is true in 3 cases:

1) $\vec{F} = \langle P, 0, 0\rangle$
2) $\vec{F} = \langle 0, Q, 0\rangle$
3) $\vec{F} = \langle 0, 0, R\rangle$

Let's prove $(*)$ in case 1) assuming that S is a graph over the xy-plane:



$$\nabla\times\vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ \partial/\partial x & \partial/\partial y & \partial/\partial z \\ P & 0 & 0 \end{vmatrix}$$

$$= \langle 0, P_z, -P_y\rangle$$

$$\text{RHS} = \iint_S \nabla\times\vec{F}\cdot d\vec{S}$$

$$= \iint_D \underbrace{\langle 0, P_z, -P_y\rangle}_{\nabla\times\vec{F}} \cdot \underbrace{\left\langle -\frac{\partial z}{\partial x}, -\frac{\partial z}{\partial y}, 1\right\rangle}_{d\vec{S}} dx\,dy$$

$$= \iint_D \underbrace{\left(-P_z\frac{\partial z}{\partial y} - P_y\right)}_{-\frac{\partial}{\partial y}(P(x,y,z(x,y)))} dx\,dy$$

Chain rule

Green's theorem

$$= \int_{C'} P(x,y,z(x,y))\,dx \;(+\,0\,dy)$$

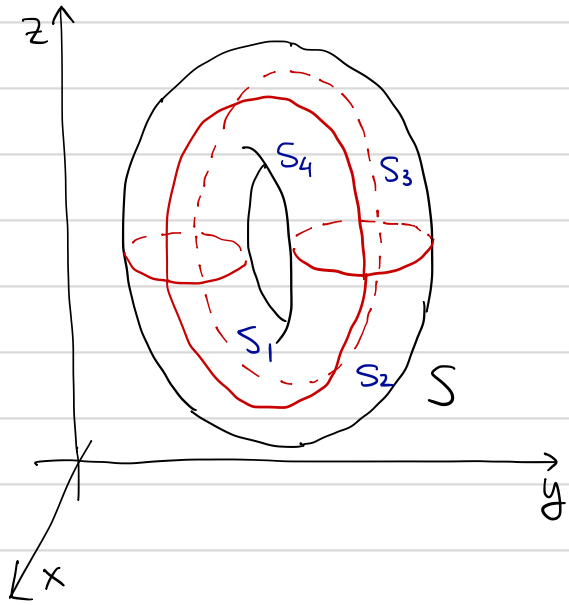
$$= \int_C P(x,y,z)\,dx$$

$$= \int_C \vec{F}\cdot d\vec{r} = \text{LHS} \;//$$

This proves $(*)$ in case 1), when S is a graph.

In most cases where S is not a graph, we can break it into pieces which are graphs over the xy-plane.

<u>Example</u>: S = Dougnut



Here we cut S along the curves where the tangent plane is vertical (contains $\hat{k}$).